

COM 769 – Scalable Advanced Software Solutions

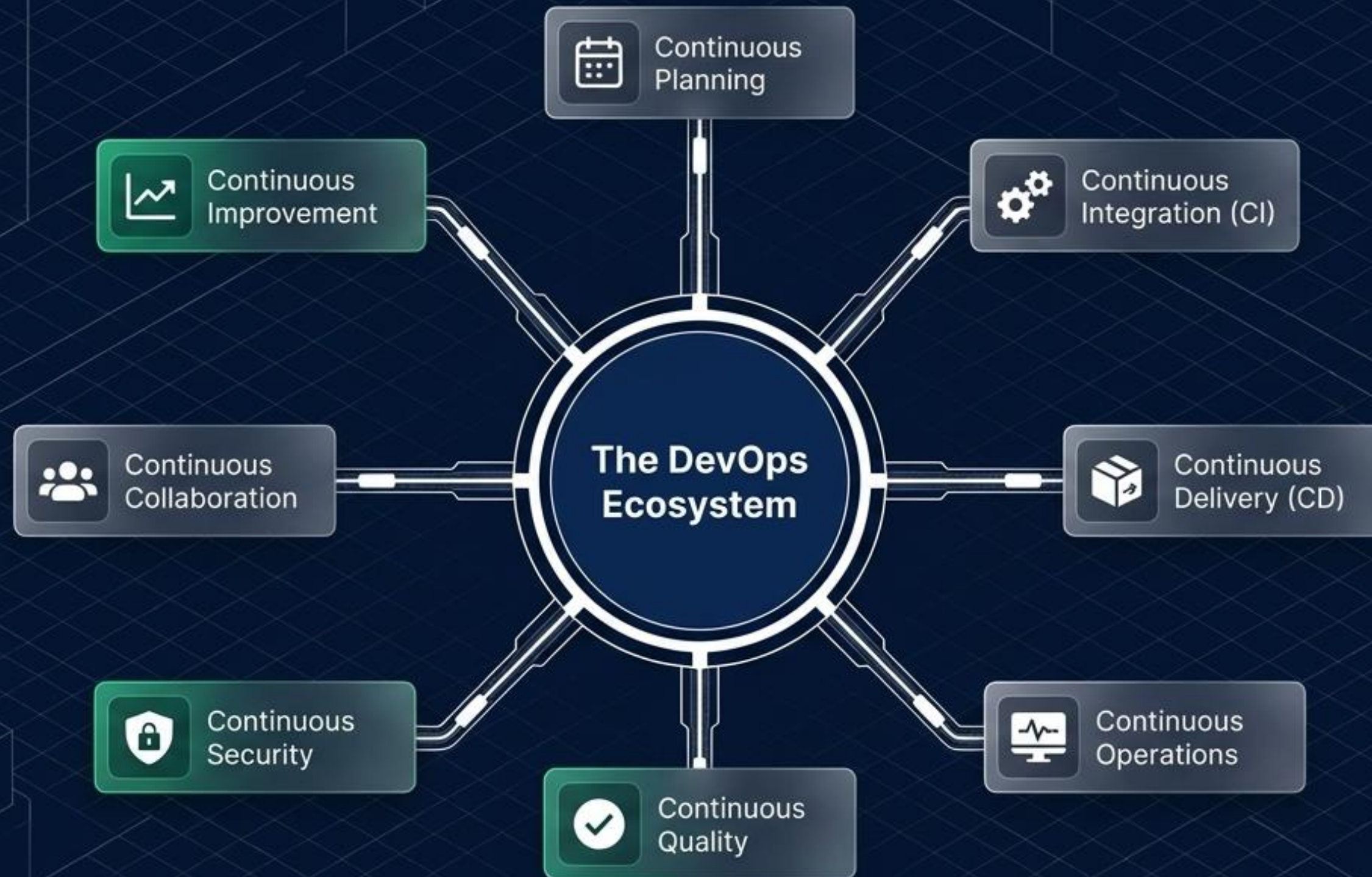
DevOps & CI/CD Pipelines

From Agile Mindsets to Automated Delivery

Presenter: Joe Rafferty







Adopting this culture allows teams to rapidly respond to customer needs, increase confidence in builds, and achieve business goals faster.

Plan

Defining features, managing backlogs, and running daily Scrums.



Develop

Version control, parallel coding, reviewing, and integrating.



**Phases are cyclical, not role-specific.
True DevOps requires every role to
be involved in every phase.**

Maintaining, monitoring, and telemetry in the cloud.

Operate



Consistent, reliable deployment to production environments.



Deliver

The Agile Mindset: Reimagining the 'Plan' Phase

What Agile Is

- + A pragmatic mindset
- + Incremental delivery
- + Continual learning
- + Team collaboration
- + Open to compromise

What Agile Isn't

- A rigid set of software tools
- A single universal framework
- Traditional waterfall planning up front

Agile is the adjective that describes a team's ability to quickly adapt to changing circumstances uncovered during the software lifecycle.

Anatomy of an Agile Sprint



The Sprint (1-4 Weeks)

Executing work in small, usable increments

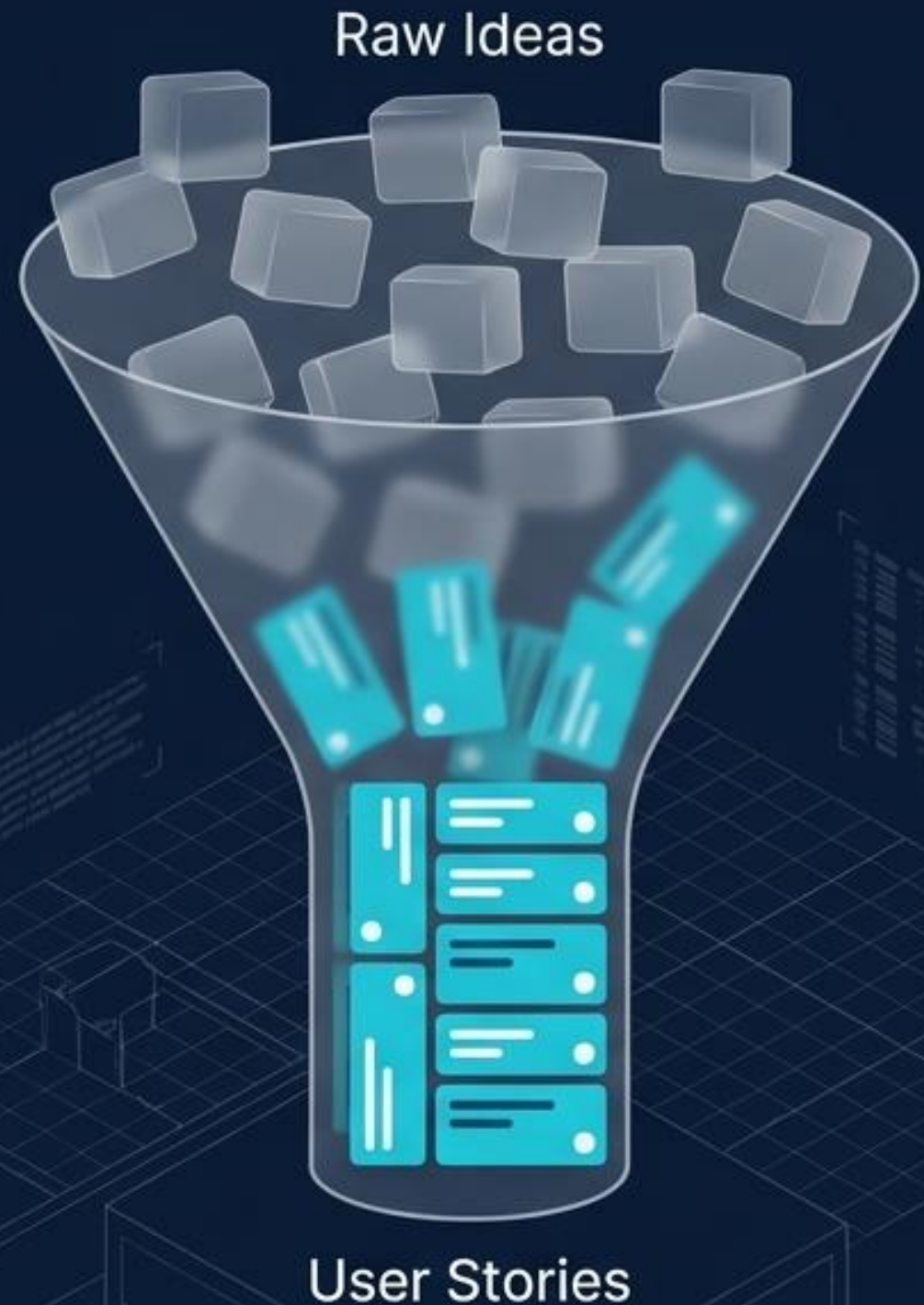
The Daily Scrum:

A 15-minute daily sync for the team to report yesterday's progress, today's plan, and blocking issues. Aided by Task Boards and Sprint Burndown charts.

The Golden Rule

All coding, testing, and quality verification must be done within each sprint.

Mastering the Backlog



The Product Owner: Owns, adds, changes, and prioritises user stories based on customer needs.

Backlog Refinement (3 Rules)

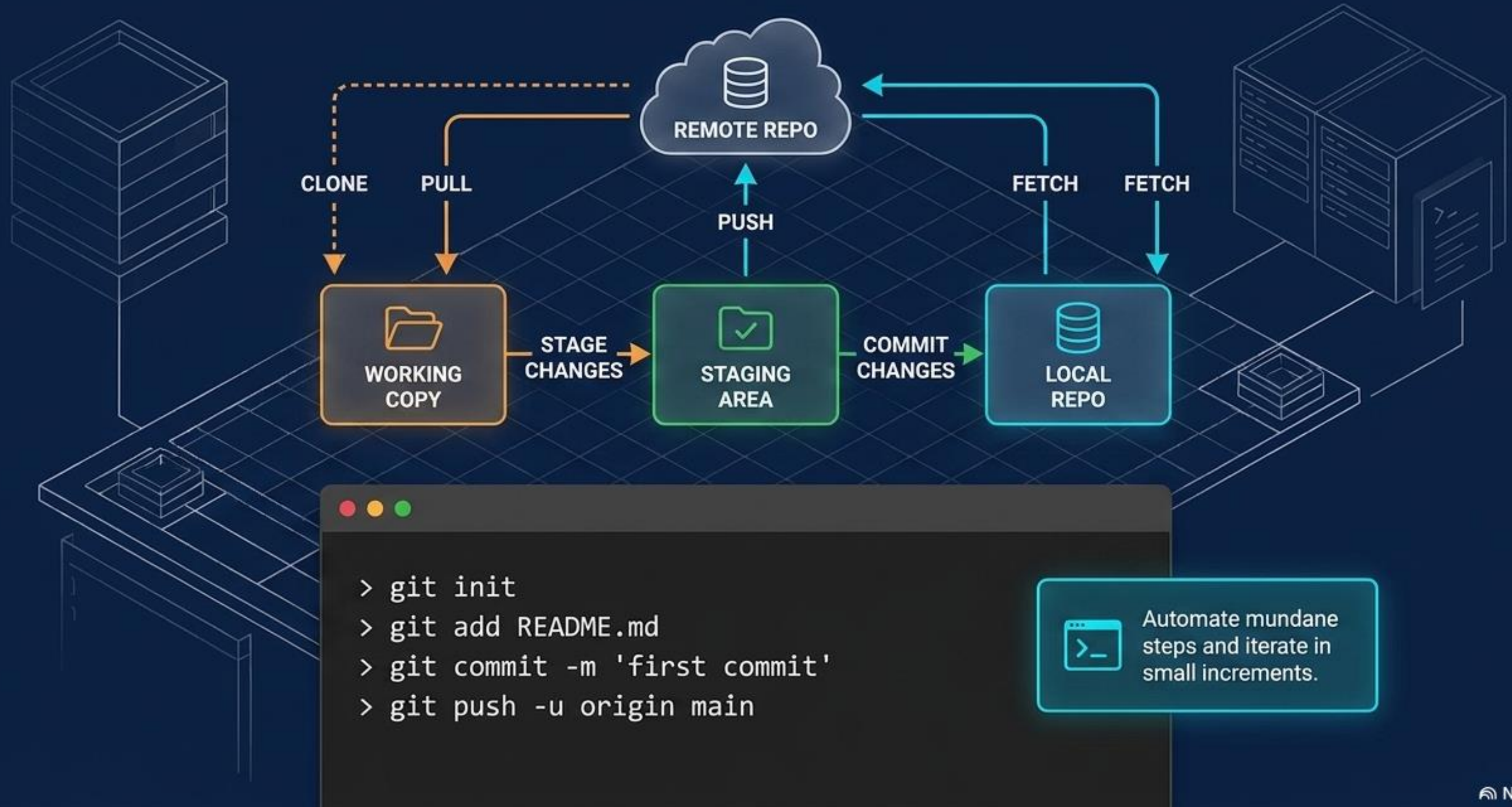
1. Look Ahead: Diligent owners refine stories 2-3 sprints in advance to account for design iterations.

2. Team Consensus: A story isn't refined until the team agrees it's ready. Day 1 surprises are a major red flag.

3. Embrace Ambiguity: Leave lower-priority items vague; focus intensely on the top of the stack.

The 'Develop' Phase: Parallel Collaboration with Git

Teams use version control to innovate rapidly and work in parallel without sacrificing stability.



Continuous Integration (CI): Integrate Early & Often



What is CI?

Automating the build and testing of code every time a commit is made to version control.

The Goal

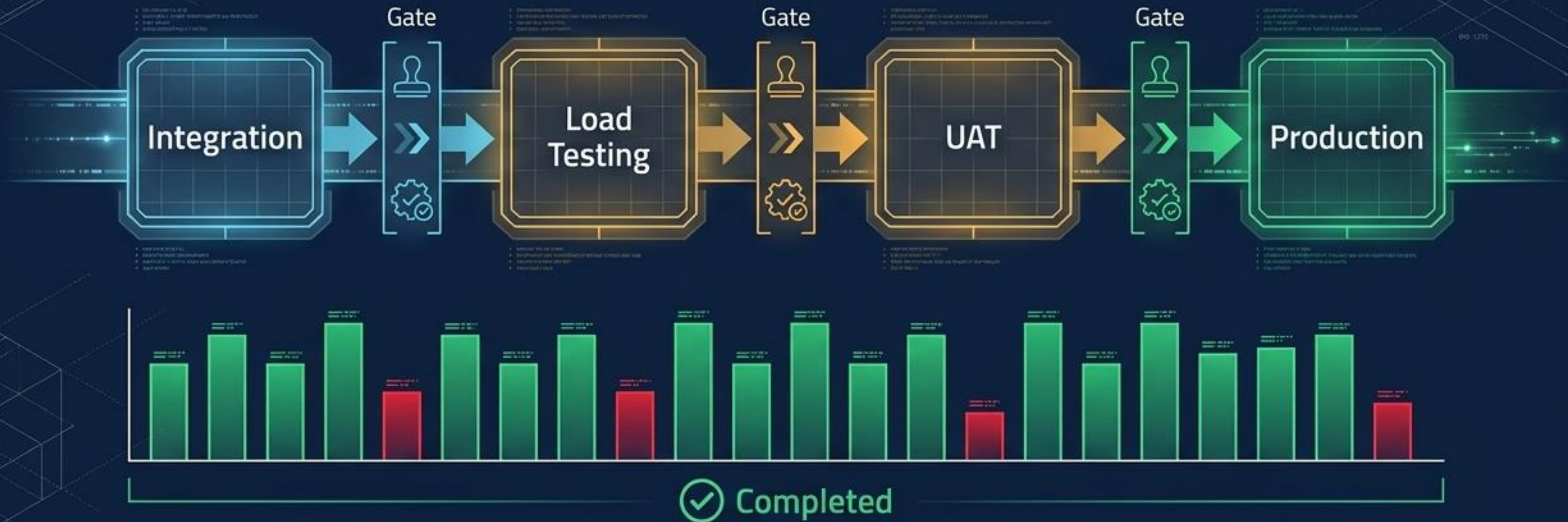
Encourage developers to merge small, frequent changes to a shared main/trunk branch.

The Trigger

Committing code automatically forces the system to grab the latest code, build it, test it, and validate the full branch.

Continuous Delivery (CD): The Release Pipeline

What is CD? Deploying applications and configuring infrastructure consistently and reliably.



The Pipeline:

Multiple staging environments support progressively longer-running activities (integration, load, and user acceptance testing).

The Gates:

Automated and manual approval gates ensure frequent, controlled, and scalable releases. Success in one environment triggers staging for the next.

The "Operate" Phase: Telemetry & Visibility

The Goal

Ensure system reliability, high availability, and zero downtime in public/hybrid clouds, alongside robust security and governance.



The Method

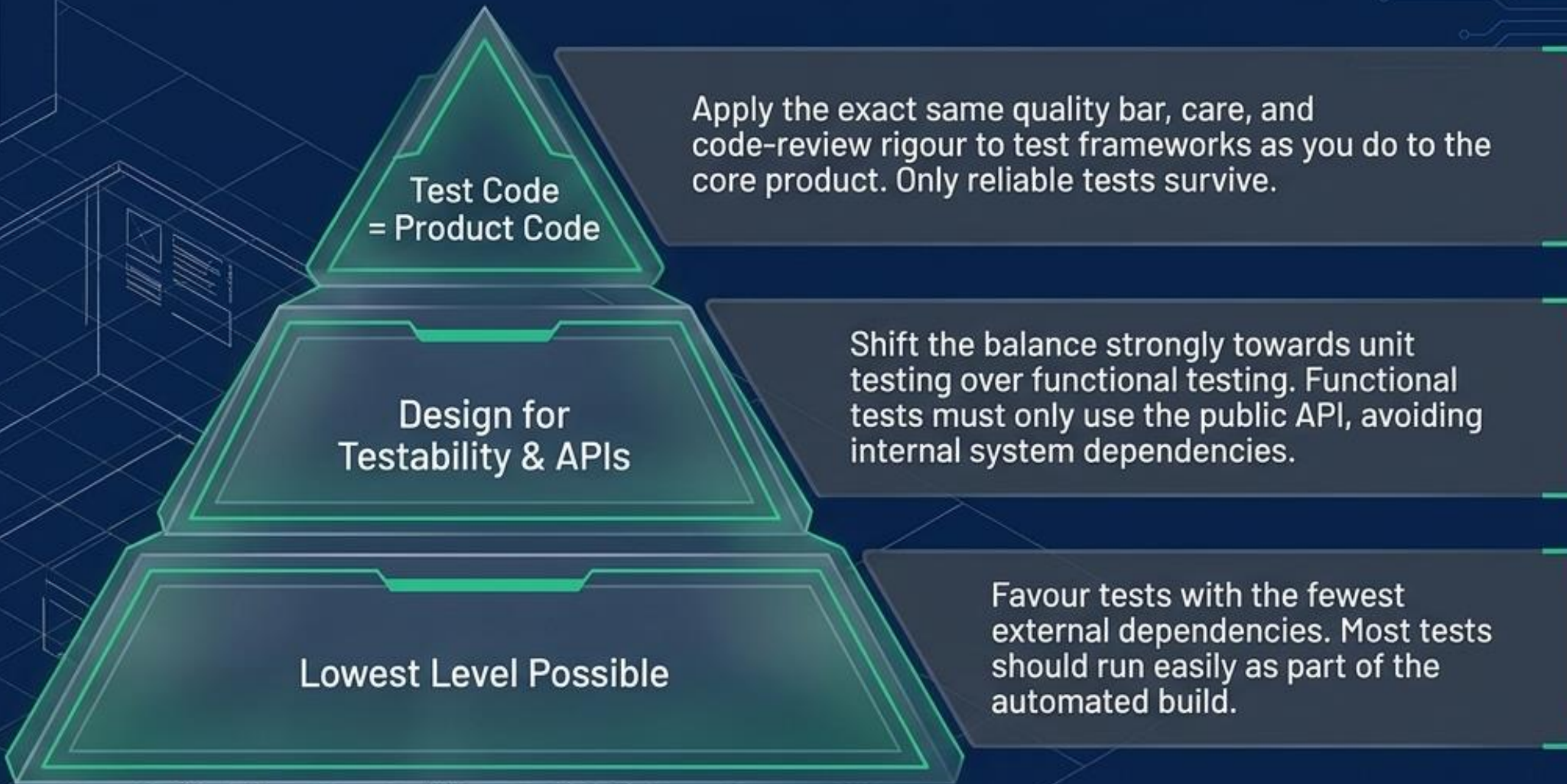
Employ safe deployment practices to catch and mitigate issues before they affect the customer experience.

The Requirement

Maintaining vigilance requires rich telemetry, actionable alerting, and full visibility into both the application and the underlying system.



Embedding Quality: Core Testing Principles



The DevOps Engine in Motion



Synthesis

DevOps is not a linear path; it is an engine. The telemetry and lessons learned in 'Operate' feed directly back into the Agile Backlog of 'Plan', continuously starting the loop over again.

Final Takeaway

Automate the mundane, iterate in small increments, and deliver continuous value.